ELSEVIER

Computers and Geotechnics, Vol. 21, No. 1, pp. 21–39, 1997
© 1997 Elsevier Science Ltd. All rights reserved
Printed in Great Britain
PII: S0266-352X(97)00001-3 0266-352X/97 $17.00+0.00

An Elastoplastic Constitutive Relation for the Soil–Structure Interface Under Cyclic Loading

I. Shahrour & F. Rezaie

Laboratoire de Mécanique de Lille, Ecole Centrale de Lille,
B.P. 48, 59 651, Villeneuve d'Ascq cedex, France

(Received 4 January 1996; revised version received 17 January 1997;
accepted 24 January 1997)

ABSTRACT

This paper includes a presentation of cyclic interface tests performed with smooth and rough surfaces on loose and dense silica sand under either a constant value of the normal stress or a zero normal displacement. These tests are then used for the elaboration of an elastoplastic constitutive relation based on the bounding surface with two families of surfaces. The constitutive relation is checked over cyclic tests performed with rough interface on both loose and dense sands. © 1997 Elsevier Science Ltd.

INTRODUCTION

Analysis of the soil-structure interaction is of major interest in geomechanics. It covers a wide range of problems such as those raised by the calculation of piles, shallow foundations, retaining walls, tunnels, embankments and reinforced earthworks. Previous work on the modelling of the soil-structure interaction was mainly intended for monotonic loading [1–14].

For structures submitted to repetitive loading like those resulting from earthquakes, ocean waves, moving traffic and machine-induced vibration, it is important to clearly describe the behaviour of the soil–structure interface under cyclic loading. Consideration of this loading in the interface behaviour suffers from a lack of experimental data necessary for both the development and verification of constitutive relations [15–20].

In this paper we present results of interface cyclic tests performed under various conditions: smooth and rough surfaces, loose and dense sands, under

either a constant value of the normal stress or a zero normal displacement. These tests are then used for the elaboration of an elastoplastic constitutive relation for modelling the cyclic behaviour of the soil–structure interface.

EXPERIMENTAL RESULTS

Eperimental device

Experimental tests were performed using a modified direct shear box. A description of this apparatus as well as preliminary tests for its verification is given in [21] and [22]. The lower part of the shear box was replaced by a block representing the structure material. Tests were performed using a polished steel block (smooth interface) and a steel block with sand grains glued at its surface (rough interface). These tests were carried out with a silica sand (Hostun sand). The particle-size distribution is given in Fig. 1. Its friction angle as measured from triaxial tests [23] varies from 38 to 41° for dense sand and from 33 to 35° for loose sand.

Monotonic tests

In order to determine the main properties of the soil–structure interface used in this work, a series of monotonic tests at constant values of normal stress (Σ_n) was performed. It included a study of the influence of both sand density and surface roughness on the behaviour of the interface.

Figure 2 shows results obtained with a rough surface and dense sand ($ID = 90\%$). It can be observed that the shear stress (Σ_t) increases with the relative shear displacement (E_t) until it reaches a peak and then decreases to

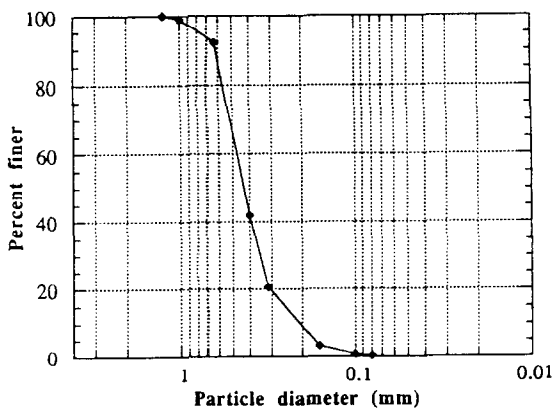

Fig. 1. Particle-size distribution of Hostun sand.

a limit shear stress. A normal displacement (E_n) accompanies the shear stress. The evolution of this displacement includes three phases (i) contraction at the very beginning of loading (ii) dilation until a relative displacement ranging from 2 to 4 mm occurs and, finally, (iii) contraction until the end of the test. These results agree with those obtained by [24]. Results with a loose sand ($ID = 15\%$) are given in Fig. 3. It can be observed that shear stress does not include softening and the normal displacement (E_n) is contracting all through the test.

Results of tests with a smooth interface are given in Figs 4 and 5. It can be observed that for dense sand, shear stress increases sharply at the beginning

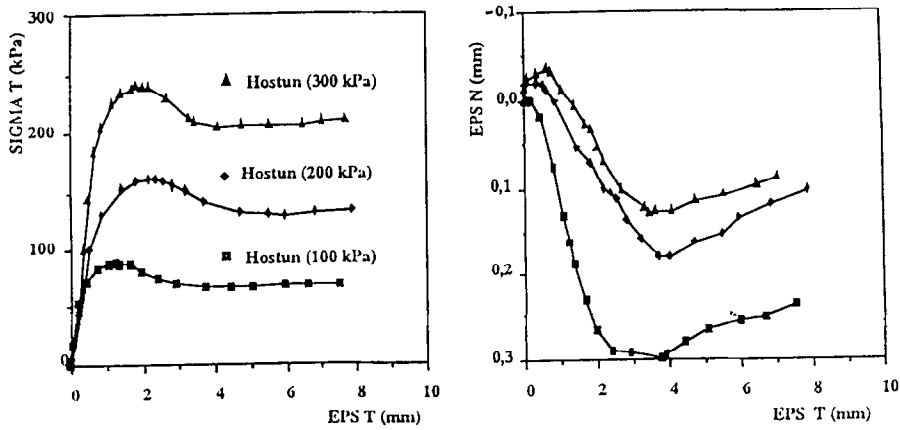

Fig. 2. Test performed at constant values of the normal stress (rough interface, dense sand).

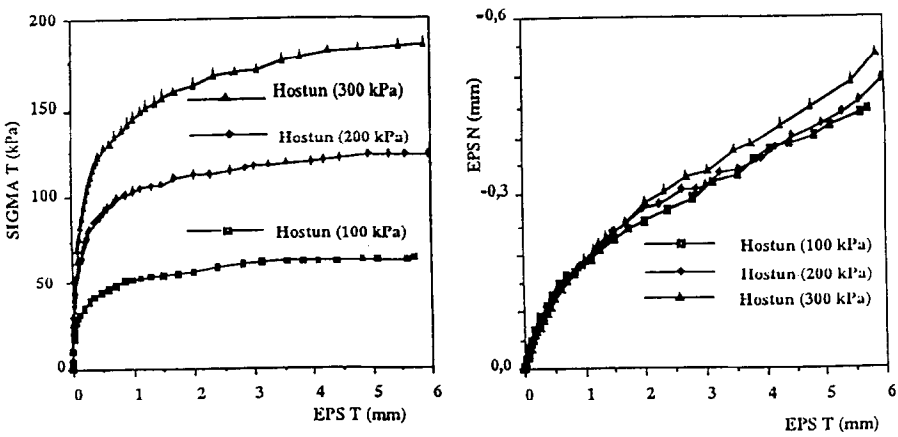

Fig. 3. Test performed at constant values of the normal stress (rough interface, loose sand).

of loading, then it decreases slightly before stabilization. In the evolution of the normal displacement (E_n), we observe contraction at the very beginning of loading followed by dilation up to a displacement ranging from 2 to 4 mm, after which we note stabilization. For loose sand we note a behaviour which is qualitatively similar to that observed with the rough surface but with lower values of both interface resistance and normal displacement.

From these tests, both frictional and dilatancy angles were determined. Their values are given in Table 1. It can be observed that the friction angle depends on both sand density and interface roughness. The friction angle of the rough interface is close to that of the sand, while for loose sand the friction angle is about 75% of that measured with the rough interface.

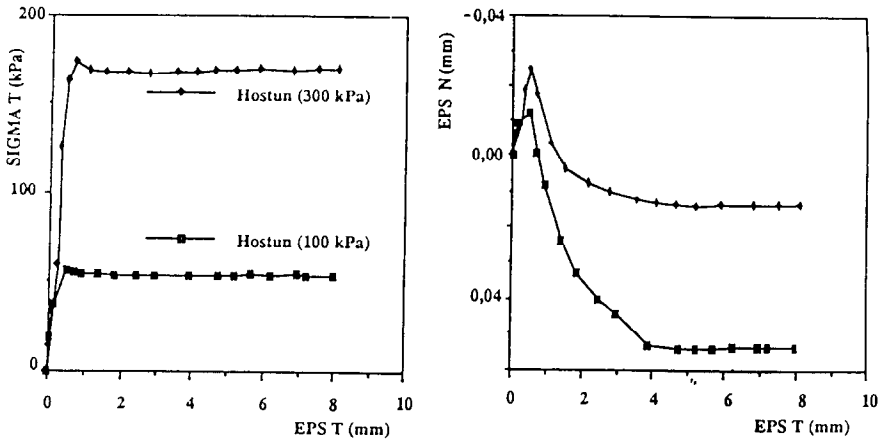

Fig. 4. Test performed at constant values of the normal stress (smooth interface, dense sand).

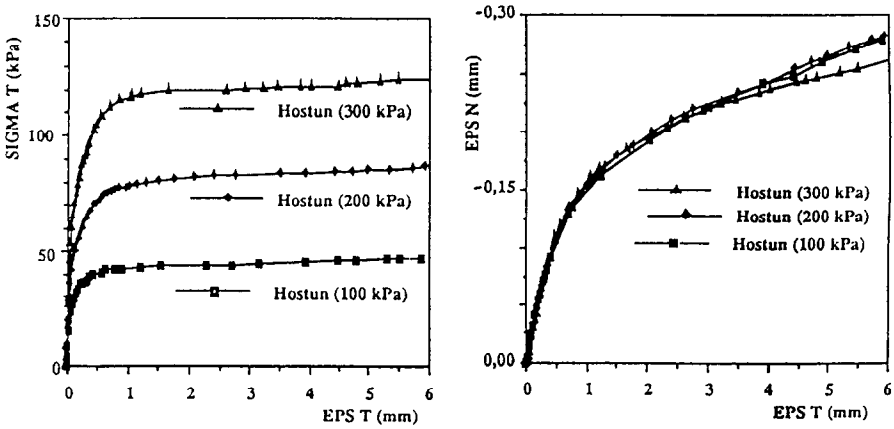

Fig. 5. Test performed at constant values of the normal stress (smooth interface, loose sand).

The angle of dilatancy (slope of the dilating part in the plane (E_n, E_t)) depends on the surface roughness. For the smooth surface, it varies from 1 to 2° while for the rough surface it varies from 4 to 5°.

Cyclic tests

The experimental program included tests with both smooth and rough surfaces on both loose and dense sands under either a constant normal stress or

TABLE 1
Friction angle of the soil-structure interface

Interface	*Dense sand ($I_D = 90\%$)*	*Loose sand ($I_D = 15\%$)*
Rough	38–40°	30–32°
Smooth	28–30°	22–24°

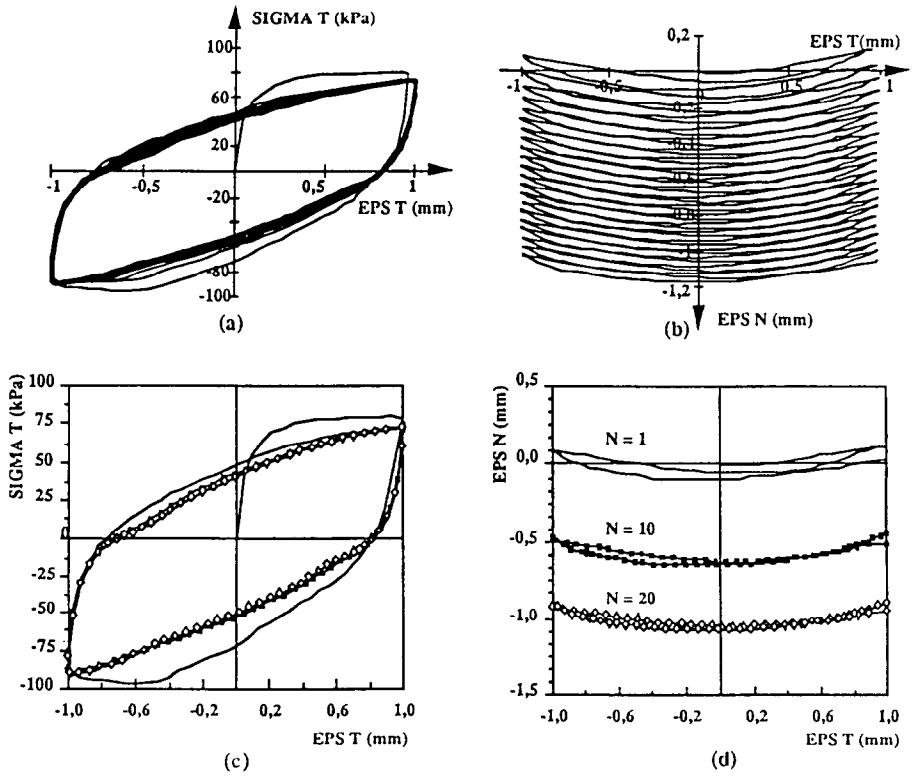

Fig. 6. Alternate interface cyclic test at constant normal stress (rough surface, dense sand). (a) Evolution of shear stress, (b) evolution of normal displacement, (c) and (d) shear stress and normal displacement at cycles number 1, 10 and 20.

a zero normal displacement. All tests were performed with an initial normal stress $\Sigma_n = 100$ kPa.

Tests at a constant normal stress

Figure 6 shows results of an alternate cyclic test with controlled shear displacement (-1 mm $\leqslant E_t \leqslant 1$ mm) performed with a rough surface on a dense

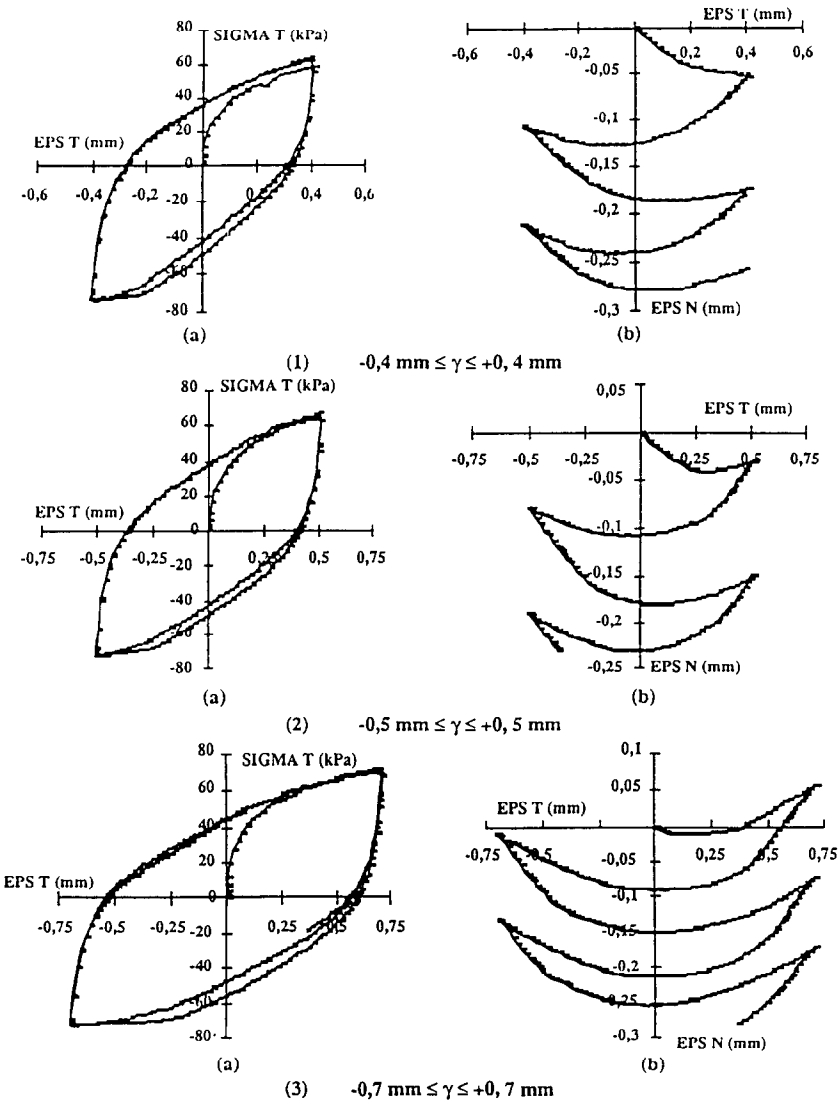

Fig. 7. Cyclic tests performed with different amplitudes (rough surface, dense sand).

sand. It can be observed that loading induces a cyclic softening (decrease of the maximal shear stress) particularly during the first cycles. Concerning the evolution of the normal displacement (E_n) we note a contracting behaviour at each stress-reversal followed by dilatation. The overall behaviour is contracting and the cumulative normal displacement per cycle decreases during loading.

The contracting behaviour of the interface at each stress-reversal is different from that observed on sand which exhibits dilatancy upon stress-unloading during triaxial tests. This difference can be attributed to the influence of the mean pressure which is constant during the shear interface test but decreases rapidly with stress-unloading during the triaxial test. In order to confirm this phenomenon, three cyclic tests were carried out with increasing amplitude (0.4 mm, 0.5 mm, 0.75 mm). Results of these tests are illustrated in Fig. 7. They confirm the contracting behaviour of the interface at each stress-reversal.

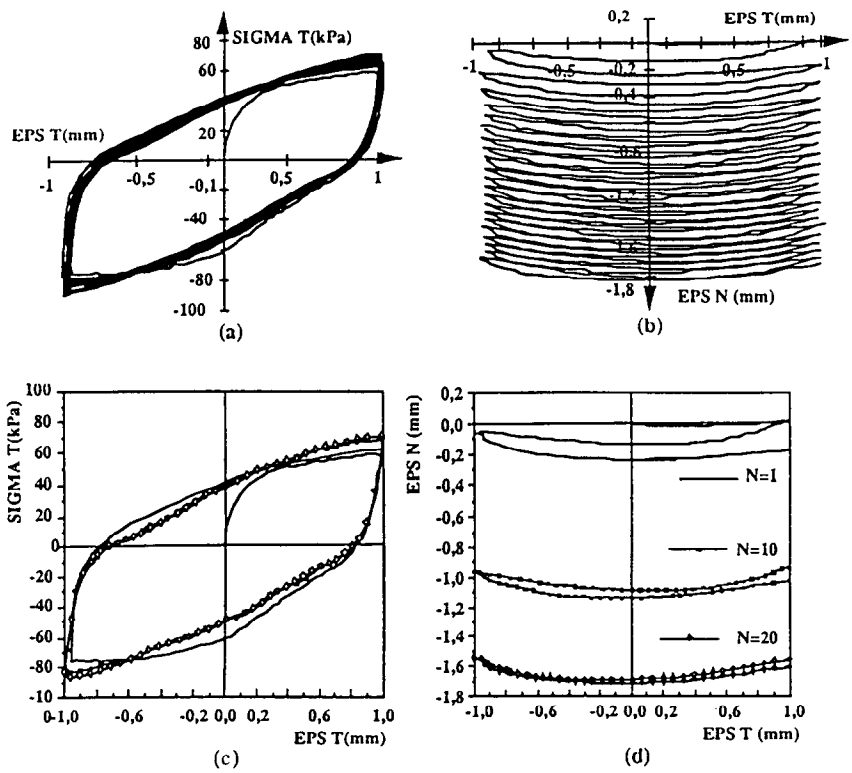

Fig. 8. Alternate cyclic test performed at constant normal stress (rough surface, loose sand). (a) Evolution of the shear stress, (b) evolution of the normal displacement, (c) and (d) shear stress and normal displacement at cycles number 1, 10 and 20.

Influence of density

Figure 8 shows the results of an alternate cyclic test (-1 mm $\leqslant E_t \leqslant 1$ mm) performed with a rough surface on loose sand. It can be observed that the cyclic loading induces an increase of the maximal shear stress (cyclic hardening). Concerning the normal displacement, we observe a contracting behaviour at each stress-reversal followed by a dilatancy which is less significant than that obtained with dense sand. The overall behaviour is contracting and the cumulative normal displacement per cycle decreases during loading. After 20 cycles, the cumulative normal displacement is around 1.7 mm while it is equal to 1.2 mm with dense sand.

Influence of the roughness of the interface

Figures 9 and 10 show results of two tests carried out with smooth interface on loose and dense sands, respectively. It can be observed that shearing induces a cyclic hardening which depends on the sand density. For dense sand we observe softening, while for loose sand we note hardening. Concerning the evolution of the normal displacement, the interface with loose sand exhibits a contracting behaviour, while with dense sand it shows a

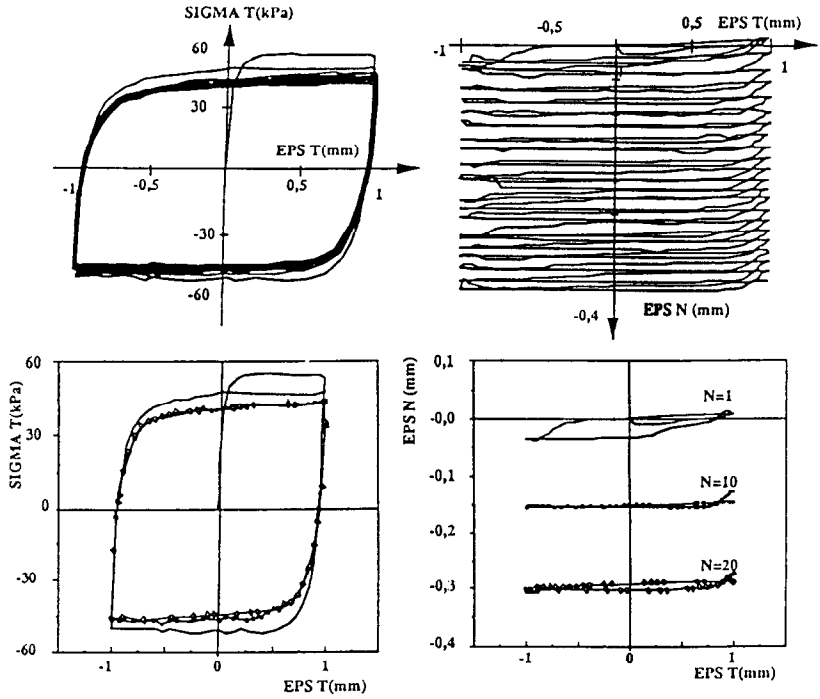

Fig. 9. Interface cyclic test performed at constant normal stress (smooth surface, dense sand).

contracting behaviour at each stress-reversal followed by a phase of quasi-stabilization of the normal displacement. The overall behaviour is contracting, particularly for loose sand. After 20 cycles, the cumulative normal displacement is equal to 0.35 mm (1 mm) for dense (loose) sand, while with the rough interface it is equal to 1.2 mm (1.7 mm) for dense (loose) sand.

Tests at zero normal displacement ($E_n = 0$)

Figure 11 shows results of an alternate cyclic test (-1 mm $\leqslant E_t \leqslant 1$ mm) performed under a zero normal displacement with a rough surface and a dense sand. It can be observed that cyclic loading induces an important variation of the normal stress. At each stress-reversal, the contracting behaviour of the interface gives rise to a decrease of the normal stress (Σ_n). This phase is then followed by an increase of Σ_n due to dilatancy. The overall contracting behaviour of the interface induces an important decrease of the normal stress and consequently an important reduction of the limit-mobilized shear, particularly during the first cycles. As the first loading induces an important decrease of Σ_n, the residual hysteresis does not exhibit symmetry about the E_t axis. With loose sand we also note a decrease of the normal stress at a rate which is higher than that observed with dense sand (Figs 11 and 12).

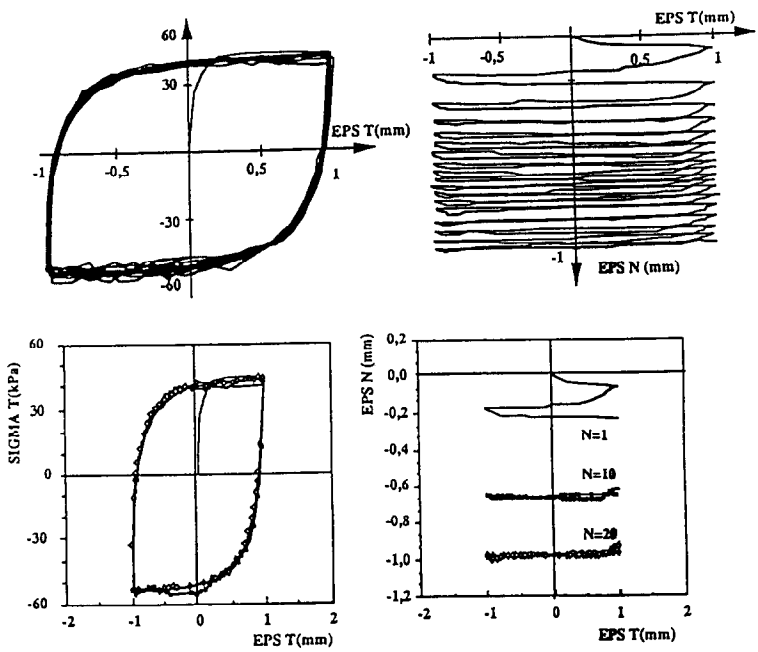

Fig. 10. Interface cyclic test performed at constant normal stress (smooth surface, loose sand).

With the smooth interface, we also observe a reduction of the normal stress and a drop of the mobilized shear stress during cyclic loading at a rate which decreases with sand density (Figs 13 and 14).

CONSTITUTIVE RELATION

Presentation of the constitutive relation

In order to describe the behaviour of the soil–structure interface observed in the precedent section, we propose a constitutive relation within the framework of plasticity using the concept of a bounding surface. This relation is

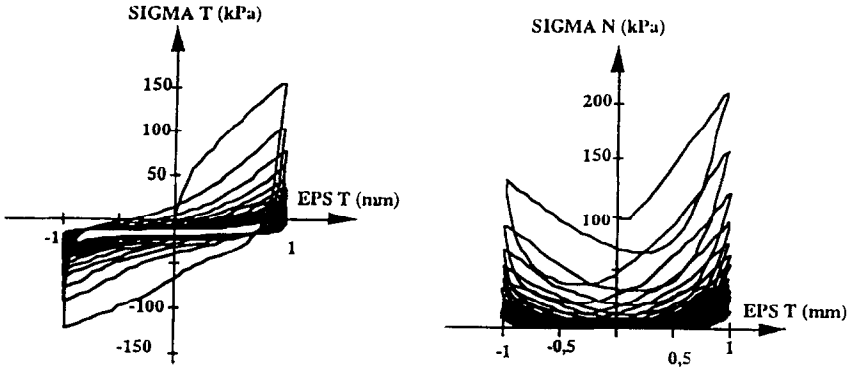

Fig. 11. Interface cyclic test performed under a zero normal displacement (rough surface, dense sand).

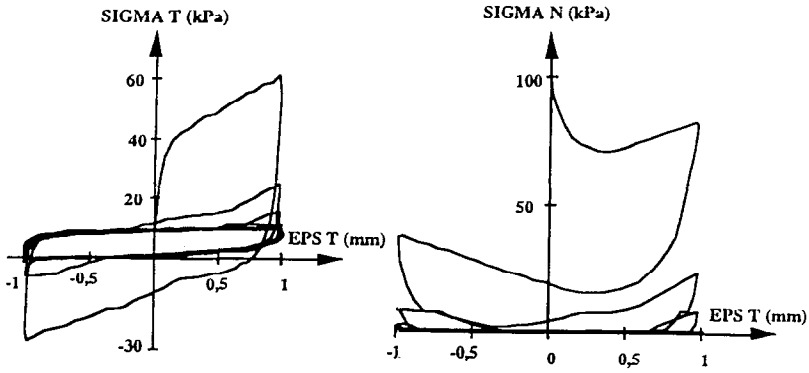

Fig. 12. Interface cyclic test performed under a zero normal displacement (rough surface, loose sand).

assumed to belong to zero-thickness interface elements. According to the theory of plasticity, the relative displacement between soil and structure (E) be decomposed into elastic (E^e) plastic (E^p) pieces.

Elastic part

The stress increment ($d\Sigma$) is linked to the elastic relative displacement increment (dE^e) by the following relation:

$$d\Sigma_n = K_n dE_n^e \qquad d\Sigma_t = K_t dE_t^e$$

K_n and K_t designate the normal and tangential elastic modulus, respectively. The former can be determined from interface shear tests performed at

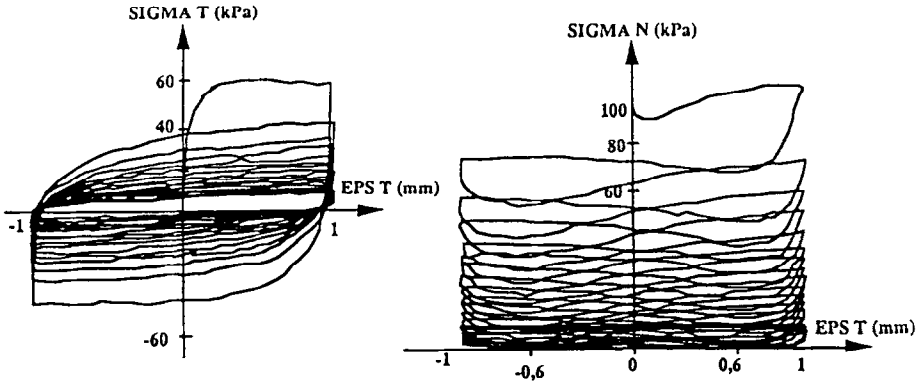

Fig. 13. Interface cyclic test performed under zero normal displacement (smooth surface, dense sand).

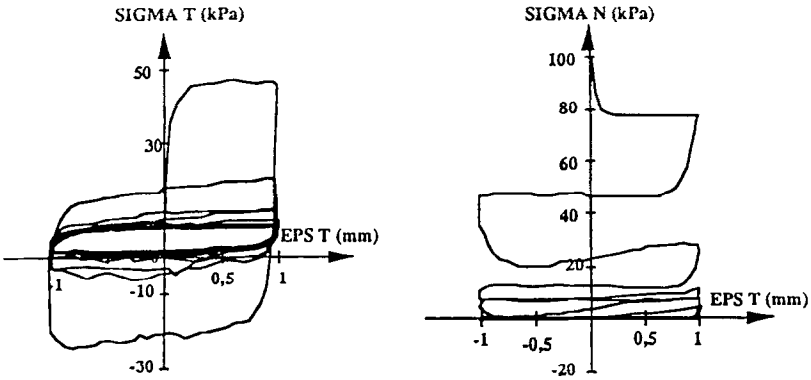

Fig. 14. Interface cyclic test performed under zero normal displacement (smooth surface, loose sand).

constant normal stress while the latter can be fixed directly from compressive interface tests or indirectly from interface shear tests performed under a zero normal relative displacement.

Plastic part

The concept of the bounding surface [25–28] is used for the description of the behaviour of the interface under cyclic loading. For the sake of simplicity, the proposed model involves only two families of surfaces (Fig. 15). The first, called the bounding surface, includes the domain of resistance of the interface. The second one, called the cyclic surface, encloses the domain of elasticity.

Yield surface

The bounding surface is assumed to be of Mohr–Coulomb type. Its expression is given by:

$$f_b = |\Sigma_t| - \Sigma_n R_{max} = 0$$

where R_{max} stands for the hardening function which depends on both friction angle and loading history:

$$R_{max} = tg\varphi + DR(1 - e^{-ADR^{*}\Gamma^{p}})$$

Γ^{p}represents the cumulative irreversible shear displacement. DR and ADR denote two constitutive parameters which can be identified from a cyclic test performed under a constant normal stress. DR presents the ratio between the amplitude of the stabilized cycle to that of the first cycle. ADR permits control of the rate of evolution of the cyclic hardening.

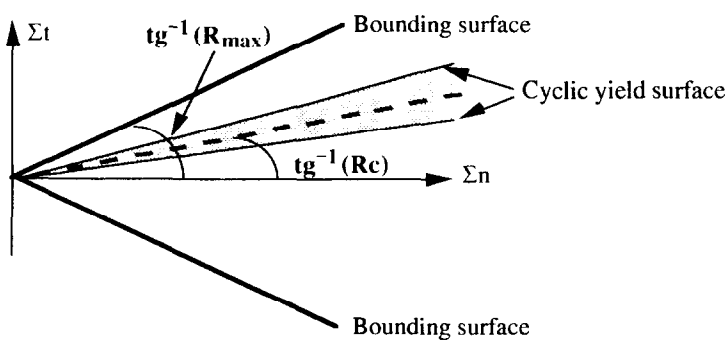

Fig. 15. Illustration of the bounding and yield surfaces.

The expression of the cyclic yield surface is given by:

$$f_c = | \Sigma_t - \Sigma_n R_c N_c | - \Sigma_n R_o = 0$$

This expression shows that the cyclic yield surface is delimited by two half-straight lines which intersect at the origin of the stress space (Σ_n, Σ_t) forming an angle referred to as '$2R_o$' which is assumed to be constant ($R_o = 0.2$). Parameter Nc denotes the direction of loading: $N_c = 1$ if the stress path is pointed in the direction of positive shear stresses, otherwise $N_c = -1$. During loading, the cyclic yield surface undergoes a movement of rotation in the stress plane. This movement is governed by the evolution of the hardening parameter R_c which stands for the angle between the axis of the cyclic yield surface and Σ_n axis. Evolution of R_c is assumed to be governed by the following relation $dR_c = \lambda H_c N_c$, where λ is the plastic multiplier. H_c is a function which controls the rate of kinematic hardening. It depends on the angle between the active branch of the cyclic yield surface and that of the bounding surface situated in the direction of the stress path:

$$H_c = \gamma_c \mid R_{max} N_c - R_c \mid^{\beta_c}$$

β_c and γ_c are two constitutive parameters which permit control of the rate of kinematic hardening. Their influence on the simulation of a cyclic test at a constant normal stress is illustrated in Fig. 16. It can be observed that, after a stress-reversal, the rate of hardening is first influenced by the parameter γ_c and then the parameter β_c. These parameters can be identified from a cyclic test as follows. Starting with $\beta_c = 1$, we determine by a process of trial and

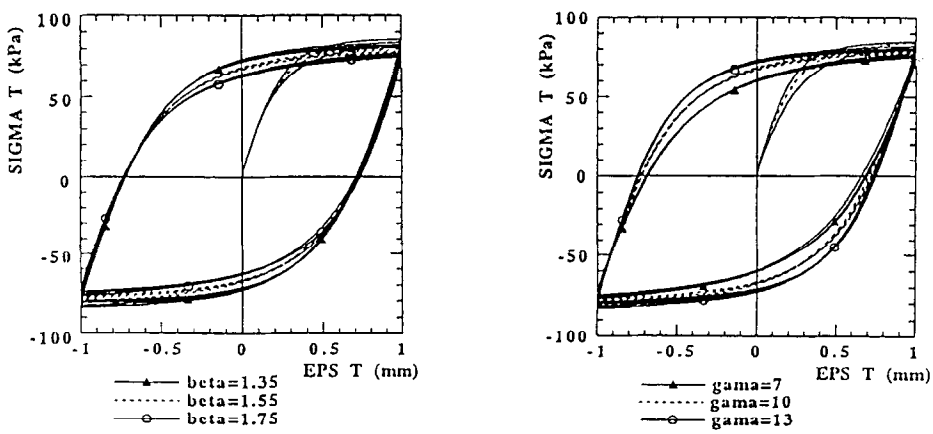

Fig. 16. Influence of parameters β_c and γ_c on the simulation of a cyclic test at a constant normal stress.

error a value of γ_c providing a good fit to experimental results. Then we seek a new value of β_c allowing improvement of the quality of fitting.

Flow rule

Cyclic tests showed that at each stress-reversal, the interface exhibits a contracting phase followed by a dilating one. In order to describe this behaviour, we adopted the following expression for the gradient of the plastic potential:

$$\frac{\partial g}{\partial \Sigma} = \begin{pmatrix} \left(tg\psi_c - \left| \frac{\Sigma_t}{\Sigma_n} \right| \right) e^{-a_c \Gamma_c^p} \\ \frac{\Sigma_t}{|\Sigma_t|} \end{pmatrix}$$

If the stress path moves to the interior of the plastic potential $\left(\frac{\partial g}{\partial \Sigma} d\Sigma < 0\right)$, the gradient of g is expressed in a local reference (related to the axis of the cyclic yield surface:

$$\frac{\partial g}{\partial \Sigma} = \begin{pmatrix} \left(tg\psi_c - \left| \frac{\Sigma_t - \Sigma_n R_c}{\Sigma_n} \right| \right) e^{-a_c \Gamma_c^p} \\ \frac{\Sigma_t}{|\Sigma_t|} \end{pmatrix}$$

where Γ_c^p denotes the plastic relative shear displacement induced since the last stress-reversal. ψ_c and a_c are two constitutive parameters. The former is identified from the locus in the plane (Σ_n, Σ_t) of points corresponding to zero normal displacement rate, the latter is intended to ensure stabilization of the normal displacement at high values of shearing. Its value can be identified from a process of trial and error.

TABLE 2

Values of constitutive parameters as determined from tests performed at a constant normal stress on rough interface

Sand	*Loose*	*Dense*
K_n (kPa/mm)	500	800
K_t (kPa/mm)	400	500
$\varphi(°)$	32	41
$\psi_c(°)$	30	32
a_c	0.01	0.03
γ_c	7	10
β_c	1.1	1.55
DR	0.1	−0.1
ADR	0.1	0.05

Finally. the proposed constitutive relation involves nine parameters: two for the elastic part (K_n, K_t), three for the bounding surface (φ, DR, ADR), two for the hardening rule (γ_c, β_c) and two for the flow rule (a_c, ψ_c). If the evolution of the bounding surface during cyclic loading can be neglected, the number of constitutive parameters is reduced to seven ($DR = ADR = 0$).

VERIFICATION

In this section we present the verification of the constitutive relation on cyclic tests performed with rough interface on both loose and dense sands.

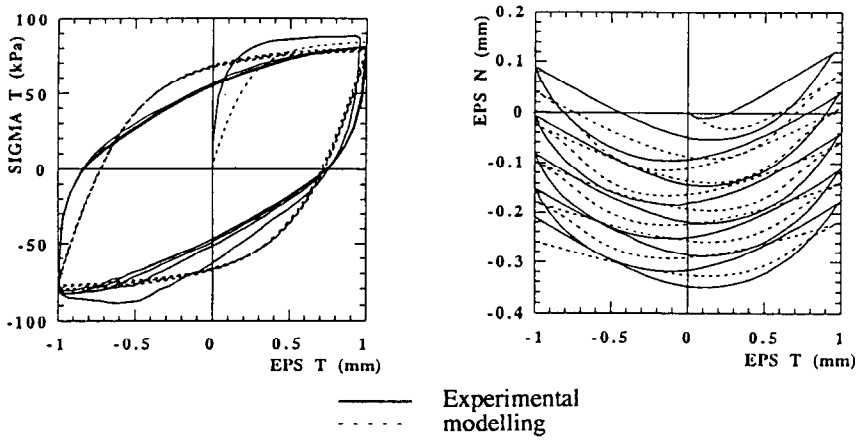

Fig. 17. Back-calculation of the test performed at constant normal stress (rough interface, dense sand).

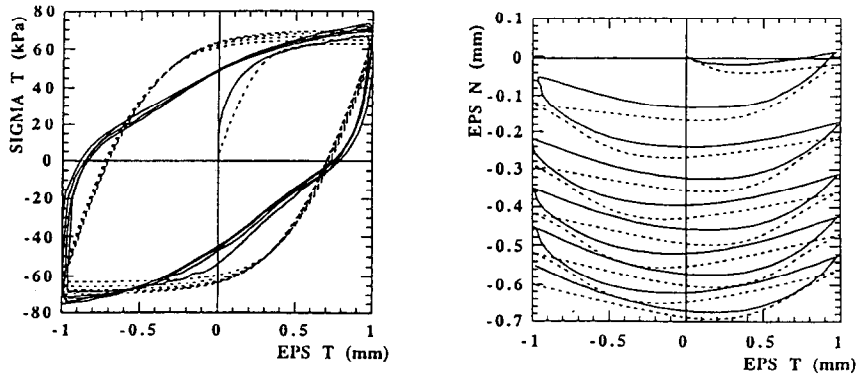

Fig. 18. Back-calculation of the test performed at constant normal stress (rough interface, loose sand).

Constitutive parameters were determined following the methodology described in the preceding section from tests performed at constant normal stress. Their values are given in Table 2. Figures 17 and 18 show results of calibration of the model on the input tests. It can be noted that the proposed relation reproduces well the evolution of cyclic hardening for both loose and dense sands, but it gives a poor fit for the first loading. An improvement of the latter requires the introduction of a supplementary constitutive parameter. Yet, for the sake of simplicity, this parameter was not retained in this version. Concerning the evolution of the normal displacement, we note that the model correctly reproduces the contracting behaviour after each stress-reversal, the dilatancy phase and the cumulative normal displacement.

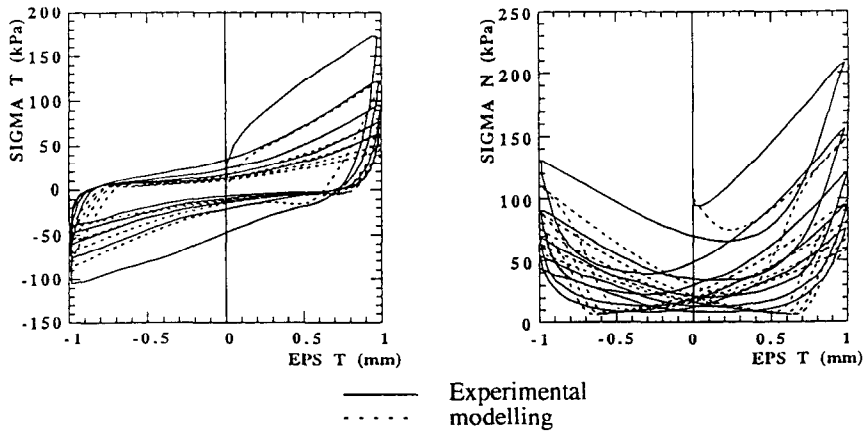

Fig. 19. Verification of the constitutive relation on test performed under zero normal displacement (rough interface, dense sand).

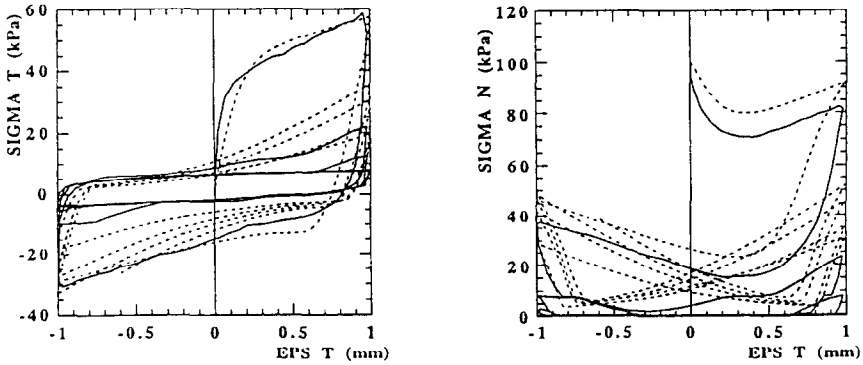

Fig. 20. Verification of the constitutive relation on test performed under zero normal displacement (rough interface, loose sand).

Figures 19 and 20 show results of the verification of the model on tests performed at a zero normal displacement. It can be observed that the model reproduces qualitatively well the evolution of the normal stress: decrease upon stress-reversal followed by an increase. At the end of loading, the model reproduces the important drop of the mobilized shear stress, particularly with loose sand.

CONCLUSION

This paper includes a presentation of both monotonic and cyclic tests performed with smooth and rough surfaces on loose and dense sands. These tests showed some important aspects of the behaviour of the interface, mainly: (i) cyclic shearing at constant normal stress induces softening (decrease of the resistance) in the case of dense sand and hardening in the case of loose sand; (ii) the interface exhibits a contracting behaviour after each stress-reversal followed by dilatancy; (iii) cyclic shearing under a zero normal displacement induces an important decrease of the normal stress and consequently a drop of the mobilized shear stress. These tests were then used for the elaboration of an elastoplastic constitutive relation based on the concept of the bounding surface with only two families of surfaces. The verification of this relation on cyclic tests gave good results.

REFERENCES

1. Boulon, M., Plytas, C. and Foray, P., Comportement d'interface et prévision du frottement latéral le long des pieux et tirants d'ancrage. *Revue Française de Géotechnique*, 1986, **35**, 311–325.
2. Boulon, M., Hoteit, N. and Marchina P., A complete constitutive law for soil-structure interfaces. *VI International Conference on Numerical Methods in Geomechanics*, Innsbruck, Balkema, Rotterdam, 1988, pp. 311–316.
3. Boulon, M. and Nova, R., Modelling of soil-structure interface behaviour — A comparison between elastoplastic and rate type laws. *Computers and Geotechnics*, 1990, **9**, 21–46.
4. Boulon, M. and Jarzebowski, A., Rate-type and elasto-plastic approaches for soil- structure interface behaviour: a comparison. *International Conference on Computer Methods and Advances in Geomechanics (IACMAC)*, Cairn, Australia, Balkema, 1991, Vol. 1, pp. 305–310.
5. Brumund, W. F. and Leonards, G. A., Experimental study of static and dynamic friction between sand and typical construction materials. *Journal Testing and Evaluation*, 1973, **1**, 162–165.
6. Desai, C. S., Zaman, M. M., Lightner, J. G. and Siriwardane, H. J., Thin-layer element for interfaces and joints. *International Journal for Numerical and Analytical Methods in Geomechanics*, 1984, **8**, 19–43.

7. Ghaboussi, J., Wilson, E. L. and Isenberg, J., Finite element for rock joints and interfaces. *Journal of the Soil Mechanics and Foundations Division*, 1973, **99**(SM10), 833–848.
8. Goodman, R. E., Taylor, R. L. and Brekke, T. L., A model for the mechanics of jointed rock. *Journal of the Soil Mechanics and Foundations Division*, 1968, **94**(SM3), 637–660.
9. Jewell, R. A. and Worth, C. P., Direct shear tests on reinforced sand. *Geotechnique*, 1987, **37**(1), 53–68.
10. Kishida, H. and Uesugi, M., Tests of the interface between sand and steel in the simple shear apparatus. *Geotechnique*, 1987, **37**, 45–52.
11. Meimon, Y., Thomas, P. A., Naudin, J. C. and Perol C., Calculation of the soil structure contact of jack up foundations. *International Conference on Numerical Methods for Coupled Problems,* Swansea, 1981, pp. 643–654.
12. Ottosen, N. S. and Olsson, K. G., Hardening/softening plastic analysis of adhesive joint. *Journal of Engineering Mechanics*, 1988, **114**(1), 97–116.
13. Potyondy, J. G., Skin friction between various soils and construction materials. *Geotechnique*, 1961, **11**(4), 339–353.
14. Wernick, E., Skin friction of cylindrical anchors in non cohesive soils. *Symposium on soil reinforcing and stabilizing techniques*. Sydney, Australia, 1978, pp. 201–219.
15. Aubry, D., Modaressi, A. and Modaressi, H., A constitutive model for cyclic behaviour of interfaces and its application in dynamic analysis. Euromech 248, Grenoble, 1989, pp. 136–139.
16. Desai, C. S., Drumm, E. C. and Zaman, M. M., Cyclic testing and modelling of interfaces. *Journal of Geotechnical Engineering*, 1985, **111**(6), 793–815.
17. Desai, C. S. and Nagaraj, B. K., Modelling for cyclic normal and shear behaviour of interfaces. *Journal of Engineering Mechanics ASCE*, 1988, **114**(7), 1198–1217.
18. Janardhanam, R. and Desai, C. S., Interface shear stiffness under cyclic loading. Soil Dynamics Earthquake Engineering Conference. Southampton, 1982, pp. 131–146.
19. Shahrour, I. and Bencheikh, B., Analysis of the soil-structure interaction under monotonic and cyclic loading. First European Conference on Numerical Methods in Engineering, Brussels, 1992, pp. 269 275.
20. Zaman, M. M., Desai, C. S and Drumm, E. C., An interface model for dynamic soil-structure interaction. *Journal of Geotechnical Engineering Division*, 1984, **110**(4), 1257–1273.
21. Rezaie, F., Etude expérimentale et modélisation du comportement de l'interface sols-structures sous chargements monotone et cyclique. Thése de doctorat, Université des Sciences et Technologies de Lille, 1994.
22. Shahrour I. and Rezaie F., Etude en laboratoire du comportement d'interface sous chargements monotone et cyclique. 6éme Colloque Franco-Polonais de Mécanique des sols appliquée, 1993, pp. 400–408.
23. Mohkam, M., Contribution á l'étude expérimentale et théorique du comportement des sables sous chargements cycliques. Thèse Docteur Ingénieur, Université de Grenoble, 1983.
24. Boulon, M., Contribution á la mécanique des interfaces sol-structure: Application au frottement latéral des pieux. *Habilitation á diriger des recherches*, Université de Grenoble, France, 1988.
25. Dafalias, Y. F and Popov, E. P., Plastic internal variables formalism of cyclic plasticity. *Journal of Applied Mechanics*, 1976, **43**, 645–650.

26. Dafalias, Y. F., Hermann, L. R. and Denatale, J. S., The bounding surface plasticity model for isotropic cohesive soils and its application. Proceedings of the International Workshop on Constitutive Relations for Soils, Grenoble, Balkema, 1982, pp. 273–287.
27. Mroz, Z., Shrivastava, H. P. and Dubey, R. N., A nonlinear hardening model and its application to cyclic loading. *Acta Mechanica*, 1976, **25**, 51–61.
28. Mroz, Z., Norris, V. A. and Zienkiewicz, O. C., Application of an anisotropic hardening model in the analysis of elastoplastic deformation of soils. *Geotechnique*, 1979, **31**, 293–302.